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Data Analytics Bootcamp (DAB 11)
Topic: Driving Cost Efficiency in Digital
Advertising

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Executive Summary

This report evaluates Apple's digital advertising campaigns and finds that while most of them achieve relatively low cost per acquisition (CPA), certain markets such as France drive costs significantly higher, creating budget leakage. Stories advertisement type deliver the lowest CPA and strongest conversion performance, whereas Video advertisement type are the least cost efficient. Geographic analysis shows that reallocating ad spending from high-cost markets like France and Mexico to more efficient regions such as the United States, Germany, and India reduce CPA by more than half and improve cost efficiency by 2.6x.

Improvements in users targeting validation are also needed to address a 78% mismatch between intended and actual user age groups. By implementing these recommendations, Apple can expect an 18% increase in purchase conversions with the same ad budget while more advanced ROI analysis is able to be performed once revenue data is incorporated in future datasets.

Overview

Audience

This report is prepared for client: Annie Chen (Senior Marketing Manager, Apple Inc.), John (Data Analytics Manager, D Advertising Pte Ltd). It is designed to provide actionable insights that support both client and internal decision-making.

Problem

Apple Inc. invests heavily in digital advertising, but cost efficiency varies across formats and markets. Inefficient capital allocation result in higher costs and weaker returns.

Goal

The report evaluates campaign performance against two criteria: achieving at least 10% increase in expected online purchases and/or 20% reduction in cost per acquisition (CPA) for each digital advertising dollar spent.

Approach

To address this challenge, the analysis examines user behavior across digital platforms and geographies, identifies inefficiencies, and recommends strategies for reallocating advertising budget. The goal is to guide Apple's marketing team toward more cost-effective targeting of the right audiences, in the right locations, through the most efficient digital advertising channels.

Data Information

Data Source

The data is obtained from fictitious advertising performance data from Kaggle. It covers from May to August 2025, including information regarding users' profiles, advertisement campaigns, advertisement events such as clicks and purchases, as well as the budget and advertisement timings. The data was consolidated into a single dataset to ensure comparability across the various channels.

Data Cleaning

All the data records were cleaned to remove duplicates, incomplete entries and erroneous data points. Dates were standardized to a uniform format and numerical fields were validated against expected ranges. This ensured the dataset was reliable, consistent, and suitable for analysis.

Data Limitations

While the dataset provides robust insights into digital campaign performance, several limitations should be noted. Online sales data was not available, which a comprehensive digital advertising analysis is unable to be performed. The predictive modeling relied on a relatively short historical time period of four months, limiting the ability to capture longer-term seasonal trends. In addition, the low purchase rate of 0.5% limits the predictive modeling performance score as well. Audience segmentation was restricted by Kaggle, which may not fully reflect comprehensive customer behavior.

Exploratory Data Analysis

Purpose

This EDA section was carried out to discover key drivers of advertising performance, validating data quality and identify actionable patterns across different campaigns. This ensures stakeholders can trust the dataset and highlight early insights before advanced modeling stage.

Data Quality

There are approximately 1% of duplicated data which have been removed from the dataset. There are no missing data in the dataset, and dataset has been verified for reasonableness. The datatype of the various categories has been updated to the appropriate data types.

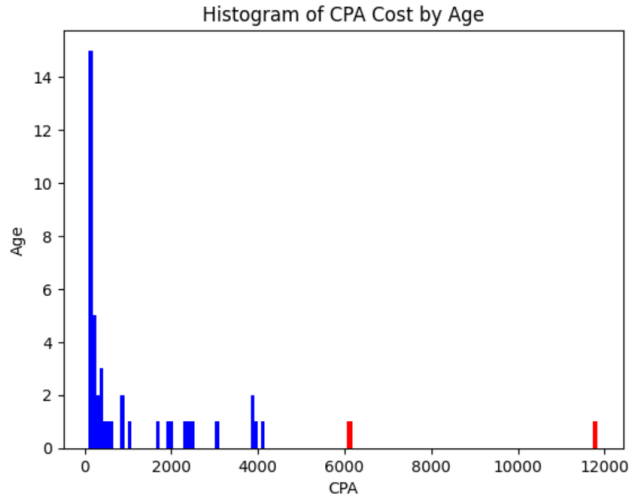
Key Insights

Figure 1. Advertising Event Types & Counts

event_type	count	proportion_%
impression	339812	84.95
click	40079	10.02
like	12013	3.00
comment	4108	1.03
purchase	2031	0.51
share	1957	0.49

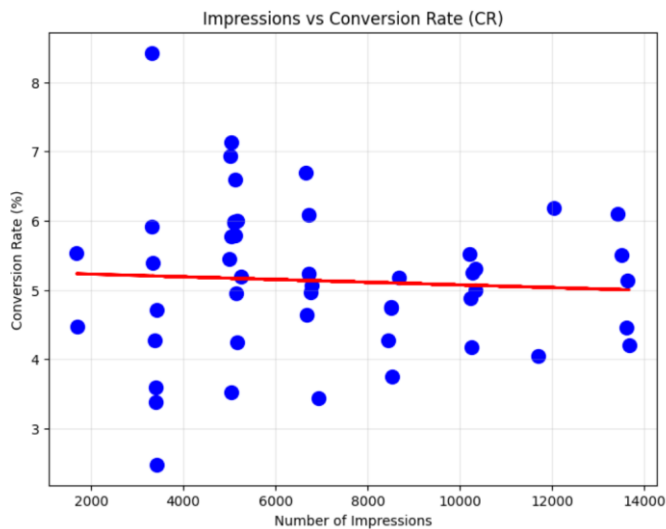
The number of impressions is the highest among all advertisement event types, making up approximately 85% of the total events (refer to above table). The second highest is number of clicks (10%) follow by number of likes (3%).

Figure 2. Histogram of CPA Cost by Age



The above table shows that the majority of the digital advertising campaigns' cost of acquisition (CPA) are low, clustering below \$4,000. However, there are a small group of online users showing significantly higher CPA values of above \$6,000. This skewed distribution suggests that while the overall digital advertising efficiency is strong, certain age group segments may require adjustment to prevent budget leakage.

Figure 3. Impressions & Advertisement Conversion Rate (CR)



As shown from above figure, scaling ad impressions does not improve conversion rates (CR). The advertising conversion (purchase) rate ranges steadily from 3-6% regardless of number of impressions. This shows that businesses should capped their digital advertising budgets at current rates.

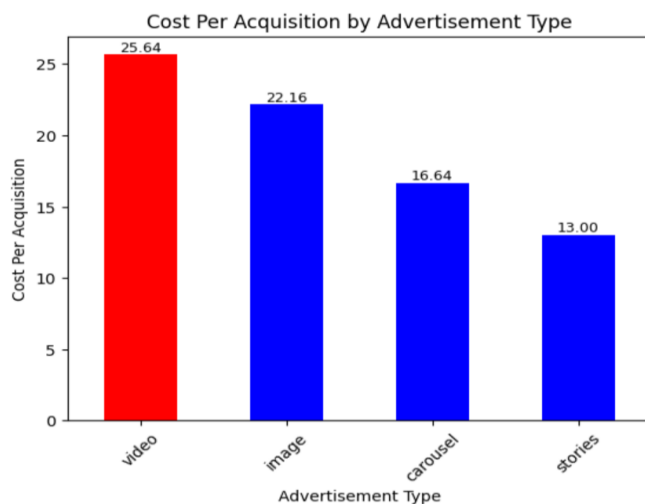
Figure 4. Relationship between Country & CR, CPA, CTR

CPA	CPA range	CTR	CTR range	CR	CR range
Low	(82.48, 692.1)	low	(9.55, 11.44)	low	(0.5, 4.26)
Medium	(692.1, 1387.53)	medium	(11.44, 12.14)	medium	(4.26, 5.83)
High	(1387.53, 18804.75)	high	(12.14, 14.52)	high	(5.83, 9.89)

country	CTR_type	CPA_type	CR_type
australia	Medium	High	Low
brazil	Low	High	Low
canada	High	Low	Low
france	Low	Medium	High
germany	High	Low	Medium
india	Low	Low	High
japan	Medium	High	High
mexico	High	Low	Medium
united kingdom	Low	Medium	Low
united states	Medium	Medium	Medium

Based on the above figure, countries such as India, Mexico and Germany have low CPA cost, medium to high conversion rate (CR). Apple should allocate more advertising budget to these countries and reduce allocation to countries such as Australia and Brazil which have high CPA cost and low CR.

Figure 5. Relationship between CPA & Advertisement Type



The median cost of acquisition (CPA) per ad type is the lowest at \$13 for Story advertisement type, and highest at \$25.64 for Video advertisement type (refer to above figure). Hence, Apple should re-allocate more advertising budget from video to story advertisement type to be more cost efficient.

Figure 6. Mismatch of Target & Actual age groups

event_type	match	click	comment	impression	like	purchase	share	CTR
0	False	31292	3211	265380	9351	1566	1521	11.79
1	True	8787	897	74432	2662	465	436	11.81

event_type	CR
0	5.00
1	5.29

There is a high proportion (78%) of mismatch between target and actual age groups of online users. This means that any age group segmenting targeting strategy is most likely ineffective, even though the conversion (CR) and click through rates (CTR) of both groups (match = event type 1, mismatch = event type 0) are similar (refer to above table).

Outliers Detection

Figure 7. Relationship between Advertising Budget & CPA per ad

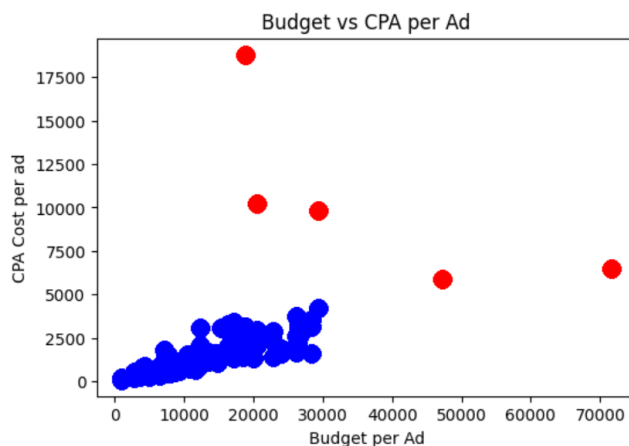
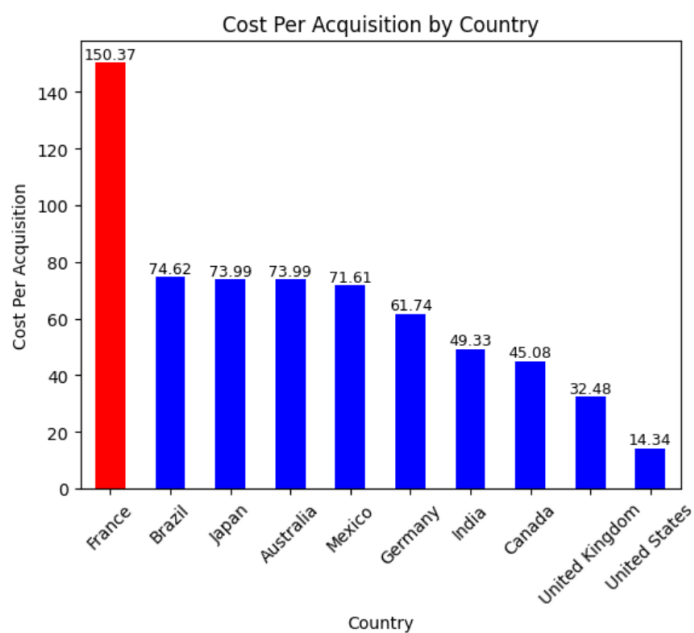


Figure 8. CPA Breakdown by Country



There is a small group of users whose CPA is significantly higher than the majority of the other online users (refer to red dots in above figure 7). This maybe due to the ad targeting at France users who have a significantly higher CPA cost as compared to other countries (see above figure 8). Hence, they are the outlier group compared to other users.

Correlation Analysis

Figure 9. Correlation between 4 categorical features

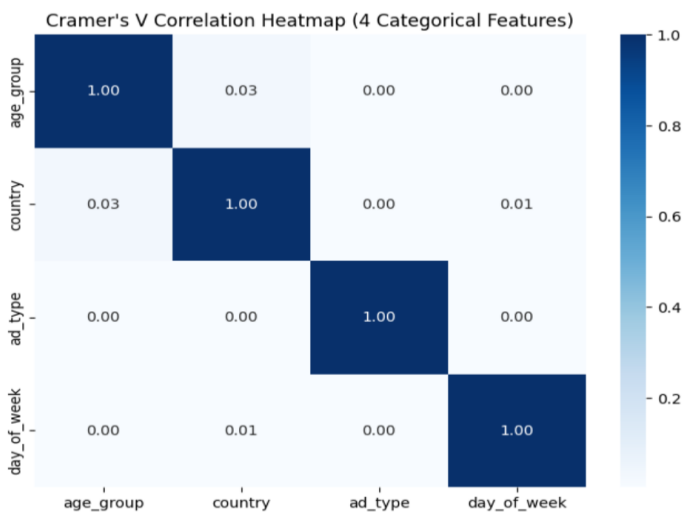


Figure 10. Correlation between Probability of Purchase & 4 categorical features

Correlation with Target Variable	
age_group	0.00
country	0.01
ad_type	0.00
day_of_week	0.00

The correlations between the 4 categorical features (age group, country, ad type, day of week) and 2 numerical features (CTR, CPA) are very low of less than 0.1. In addition, the correlation between target variable and categorical features are also very low at 0.01 (see above). This shows that there is no multicollinearity issue among the features. However, these low correlations are also likely due to the low purchase rate (class size) of 0.5%.

Predictive Modeling

Model Rationale & Drivers

The logistic regression model is being chosen due to its ability to handle large datasets that consists of a mix of categorical and numerical features, as well as a binary target (purchase) variable. In addition, the model can train and predict faster and needing lesser computing power as compared to other similar models such as random forest.

There is a second model, Light Gradient Boost Model (GBM), that is used for the predictive modeling. Despite having a higher recall score, the extremely low and imbalanced F1 score of 0.01 will result in significantly higher number of incorrect purchase predictions, hence more wastage of advertising budget for Apple. In the end, the logistic regression model with a much higher F1 score of 0.34 is chosen due to its relatively lower wastage of advertising budget.

Figure 11. Coefficients & Odds Ratio of Logistic Regression Model

Feature	Coefficient	OddsRatio
country_japan	0.36	1.44
country_mexico	0.27	1.31
w_cpa	-0.24	0.78
country_france	-0.24	0.79
day_of_week_tuesday	-0.21	0.81
day_of_week_wednesday	-0.14	0.87
age_group_25-34	0.12	1.13
country_united states	0.12	1.13
age_group_35-44	0.12	1.12
day_of_week_sunday	-0.09	0.92

Features such as Countries and CPA are the strongest drivers of Probability of Purchase as shown from above table. Businesses should increase digital advertising targeting at countries like Japan and Mexico as every unit increase in these countries will result in an increase of 36% and 27% in the probabilities of purchase on the online users respectively. In addition, businesses should reduce the CPA as an increase in CPA will reduce the probability of purchase on the online users.

Performance Metrics

Figure 12. Logistic Regression Model Performance Metrics

	precision	recall	f1-score	support
0	1.00	0.50	0.66	119391
1	0.01	0.55	0.01	609
accuracy			0.50	120000
macro avg	0.50	0.52	0.34	120000
weighted avg	0.99	0.50	0.66	120000

The logistic regression model achieves macro average performance scores (precision: 0.50, recall: 0.52, f1: 0.34, accuracy: 0.50) as per above table. These scores are not considered high, means the approximately half of the model's predictions are true. The extremely low precision score of 0.01 for the minority class (1) is due to the low purchase rate of 0.5% (imbalanced class size).

Segment Analysis

Figure 13. Expected CPA by Country

	ad_budget	expected_purchases	expected_cpa
country			
france	11930.0	65.0	183.54
japan	11953.0	112.0	106.72
mexico	11806.0	113.0	104.48
brazil	11994.0	118.0	101.64
australia	12006.0	138.0	87.00
germany	11887.0	152.0	78.20
india	11867.0	183.0	64.85
canada	11831.0	194.0	60.98
united kingdom	11938.0	283.0	42.18
united states	11941.0	617.0	19.35

The top three countries with the lowest expected cost of acquisition (CPA) are United States (\$19), United Kingdom (\$42) and Canada (\$61). The median expected cost of acquisition (CPA) of the countries segment is calculated to be \$82.60. Businesses can re-allocate their digital advertising budget to focus on those countries whose CPA are below the median CPA cost.

Scenario Analysis

Figure 14. Increasing CPA by 10% on Probability to Purchase

decile	baseline_purch_prob	scenario_purch_prob	Difference_pct
1	0.37	0.36	-3.25
2	0.45	0.45	-1.51
3	0.47	0.47	-1.16
4	0.49	0.48	-0.88
5	0.50	0.49	-0.65
6	0.51	0.50	-0.57
7	0.51	0.51	-0.49
8	0.53	0.52	-0.52
9	0.54	0.54	-0.41
10	0.57	0.57	-0.40

Increasing CPA by 10% will reduce the probability of purchase from -0.40% to -3.2% at every decile (refer to above) even though the overall mean probability to purchase is unchanged due to low purchase rate (0.5%). As such, this shows that businesses should reduce their average CPA in order to result in higher chance of purchase on their products.

Assumptions & Limitations

The above logistic regression model assumes that online users' behavior patterns are representative of the behavior of the real world's online users' population, and market conditions will remain unchanged over time. Another assumption is that the target purchase variable is only linearly related to the features of the model, which are also assumed to be independent of each other.

There are several limitations, namely the low 0.5% purchase rate reduce model accuracy, making predictions less reliable. On top of that, the four-month time period of the dataset limits visibility of seasonal or long-term digital advertising trends. In addition, the absence of revenue data of advertisements limits the coverage of the digital advertising analysis. Last but not least, the high mismatch of the target group segments limits the value of the age group segment analysis.

Recommendations

Budget Allocation

Businesses can re-allocate 20% of their digital advertising spending from video advertising type (lowest expected number of purchases) to stories advertising type (highest expected purchases). This will result in a 18% increase in expected number of purchases (conversion) with the same amount of budget.

Furthermore, businesses can also re-allocate their digital advertising budget from high CPA cost countries (France and Mexico) to countries that have high conversion rates and low expected CPA costs (United States, Germany & India). In this way, Apple should reduce their expected CPA more than 50% from \$623 to \$234, which would also significantly increase their digital advertising cost efficiency by 2.6x.

Data Improvements

Digital advertising revenue data should be included for more comprehensive analysis involving Return on Investment (ROI). In addition, longer time period data should be provided to allow for seasonality and longer-term advertising trends.

In addition, advertising companies like us should improve our target group data validation process to reduce the significant target age group mismatch issue. Consequently, businesses can deploy more effective age group segment advertising strategies.

Model Improvements

Due to the large imbalanced class size (purchase rate of 0.5%), the highest macro average precision and recall score of logistic regression model is only 52% which is not considered high. As a result, a second model, Light GBM, was being used and evaluated against the first model. Despite having a high recall score of 99%, its precision and F1 scores are very low 1% respectively.

This unbalanced recall-precision score of the Light GBM model leads to the conclusion that first model is more appropriate. More in-depth tuning of the model's hyper parameters can be performed which may improve the performance scores of both models.

Next Steps

Apple should re-allocate 20% of digital advertising budget from Video to Stories ads to capture an 18% increase in conversions. A pilot reallocation from France and Mexico to the United States, Germany, and India should be tested over one month to confirm cost efficiency gains. Targeting validation must be strengthened to reduce the 78% age-group mismatch.

The dataset should be expanded to include digital advertising revenue data and longer time coverage to enable ROI and seasonality analysis. Predictive modeling should be refined with ensemble methods to balance recall and precision. Finally, monthly dashboards should be established to track CPA, conversions, and ROI to ensure sustained impact.

Appendix

Metric Definitions

Click Through Rate (CTR) = Clicks / Impressions * 100

Conversion Rate (CR) = Purchases / Clicks * 100

Cost Per Acquisition (CPA) = Campaign Advertising Budget / Purchases * 100

Data Dictionary Table

Field Name	Definition	Data Type
event_id	Unique identifier for each event	integer
ad_id	Unique identifier for each ad	integer
user_id	Unique identifier for each user	string
timestamp	Exact date and time when the event occurred	string
day_of_week	Day of the week derived from timestamp	string
time_of_day	Time bucket representing when event occurred	string
event_type	Type of user interaction	string
campaign_id	Unique identifier for each advertising campaign	integer
ad_platform	Platform where the ad was served	string
ad_type	Format of the ad	string
target_gender	Gender intended for targeting	string
target_age_group	Age group intended for targeting	string
target_interests	Interests intended for targeting	string
name	Campaign name or label	string
start_date	Campaign start date	string
end_date	Campaign end date	string
duration_days	Total campaign duration	integer
total_budget	Total allocated budget for campaign	float
user_gender	Gender of the user who saw the ad	string
user_age	Age of the user who saw the ad	integer
age_group	Age bucket of the user	string
country	Country where ad was served	string
location	Specific location (city/region) where ad was served	string
interests	Interests of the user who saw the ad	string

Tools & Environment

This report adopts SQL (structured query language) for joining of data tables. In addition, the report use python (pandas & scikit-learn) for data cleaning, exploratory data analysis, modeling and visualization for stakeholder reporting.